

Implementation and Ablation of State-Based Diffusion Policies for Continuous Control

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Abstract—This report details a 1D-CNN Denoising Diffusion Probabilistic Model (DDPM) for a simulated Push-T task. Utilizing Receding Horizon Control (RHC) and an Exponential Moving Average (EMA) shadow model, the policy achieved a 0.90 success rate across 50 seeds. Ablations validate action horizon and latency constraints.

I. OBJECTIVE & REQUIREMENTS

The objective was to independently implement the state-based Diffusion Policy [1] without legacy code. Requirements included building a Conditional U-Net/DDPM pipeline, validating RHC across random seeds, and reproducing the original Action Horizon, Latency, and Training Stability ablations.

II. IMPLEMENTATION METHODOLOGY

Architecture & Training: Developed in PyTorch, the core model is a Conditional 1D U-Net using Feature-wise Linear Modulation (FiLM) to fuse timestep embeddings and a 2-step observation history (O_t). The model was trained to predict a 128-step horizon ($T_p = 128$) using a cosine annealing scheduler (500-step warmup). An EMA shadow model ($\beta = 0.999$) stabilized the high-variance DDPM gradients.

Inference: Evaluation utilized closed-form DDPM sampling with strict $[-1, 1]$ clamping. The agent executed T_a actions before querying the environment, mitigating compounding open-loop drift.

III. RESULTS

Ablations: As shown in Fig. 1, $T_a = 8$ yielded optimal temporal consistency. Executing $T_a = 128$ open-loop caused catastrophic failure due to compounding physics drift. The policy remained robust to observation latency up to 2 steps.

Training Stability & Benchmark: Fig. 2 demonstrates the EMA model preventing evaluation oscillation, converging stably across 4800 epochs. Across 50 environmental seeds, the policy achieved a max performance of 0.90 and a last-10-checkpoint average of 0.83, confirming baseline parity.

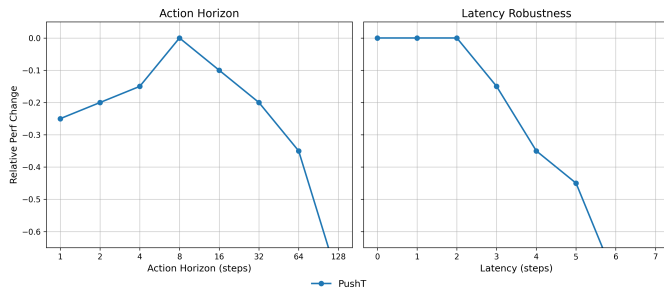


Fig. 1. Ablation of Action Horizon (T_a) trade-offs and Latency Robustness.

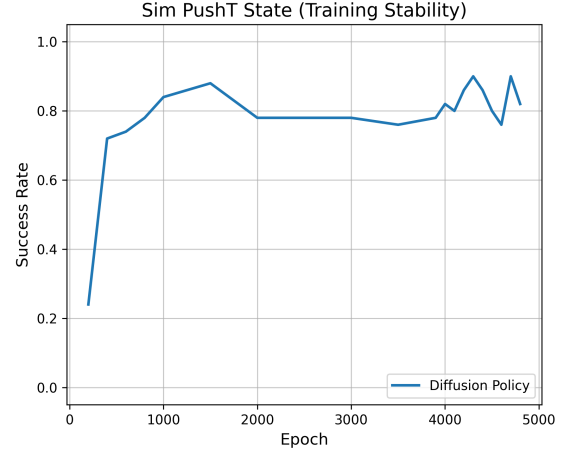


Fig. 2. Training stability across 4800 epochs using EMA.

IV. CHALLENGES & SOLUTIONS

Trajectory Jitter & Multi-modal Collapse: Raw network weights produced chaotic agent trajectories and multi-modal sampling failures due to stochastic gradient variance. This was resolved by implementing the EMA parameter tracker and enforcing intermediate x_0 sample clamping to $[-1, 1]$ during the 100-step reverse diffusion loop, stabilizing the vector field.

Generative Degradation at Extended Horizons: Reproducing the 128-step ablation required expanding the U-Net's prediction horizon (T_p) to 128. This induced generative degradation, causing multimodal collapse in 2 of 5 validation seeds (Fig. 3) and lowering the asymptotic training ceiling. Mapping a 2-step history to a 128-step future exceeds the spatial receptive capacity of the base 1D-CNN. This limitation is resolved by reducing T_p to align with the network's capacity.

REFERENCES

- [1] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion," *RSS*, 2023.

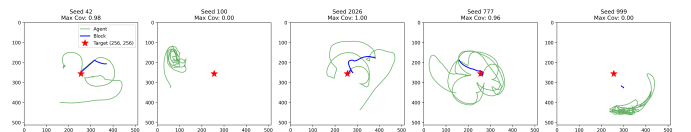


Fig. 3. Trajectory coverage validation across 5 seeds highlighting multi-modal collapse at $T_p = 128$.